

Maci Kesler

mkesler1@kent.edu

Center for Nuclear Research

Department of Physics

Kent State University

Kent, OH USA 44240

Website: macink.github.io

Education

- 2022 – Present  **Ph.D., Physics, Kent State University**
Advisor: Zhangbu Xu
Research Interests: Diffractive scattering, heavy-ion/electron-ion collisions, nuclear structure, quark-gluon plasma, vector meson production.
- 2022 – 2024  **M. A., Physics, Kent State University**
- 2019 – 2022  **B.S., Computational Physics, University of Nevada, Las Vegas**
Summa Cum Laude
Honors Thesis: *Dynamics of magnetic monopoles with charged matter.*
- 2018-2019  **A.A, Elementary Education, College of Southern Nevada**

Appointments

- 2024 – Present  **Graduate Research Assistant**, Kent State University, Kent, OH.
- 2022 – 2024  **Graduate Teaching Assistant**, Kent State University, Kent, OH.

Service

- 2026 – 2027  **City Coordinator - Pint of Science**, Northwater Brewing, Kent, OH.
- 2025 – 2026  **President - Physics Graduate Student Association**, Kent State University, Kent, OH.
 **Event Manager - Pint of Science**, Northwater Brewing, Kent, OH.
- 2025  **SciPals**, Miller South School for the Visual and Performing Arts, Akron, OH.
 **STAR Collaboration Meeting Organizer**, Kent State University, Kent, OH.
 **Fascinating Physics Tour Guide**, Brookhaven National Lab, Upton, NY.
 **Host - Physics Department Flash Talk Series**, Kent State University, Kent, OH.
- 2024 – 2025  **Organizer - Center for Nuclear Research**, Kent State University, Kent, OH.
- 2024  **STAR Shifts**, Brookhaven National Lab, Upton, NY.
 **Student Representative - Student Academic Complaint Committee**, Kent State University, Kent, OH.
 **Event Organizer - Total Solar Eclipse**, Kent State University, Kent, OH.
- 2023 – 2025  **Graduate Program and Admission Committee Representative - Physics Graduate Student Association**, Kent State University, Kent, OH.
- 2020 – 2022  **President - Society of Physics Students**, University of Nevada, Las Vegas, NV.

Computing

-  **Programming Languages:** C++, Mathematica, Python, Qiskit, ROOT, PyTorch
-  **Software:** Linux, Git/GitHub, Jupyter Notebook, VS Code, ROOT, HTCondor, Rucio

Achievements

Awards

- 2025  **Department of Energy - Office of Science: Graduate Student Research Award**, Brookhaven National Lab, Upton, NY.
- 2024  **Departmental Graduate Award: Outstanding Service Award**, Kent State University, Kent, OH.
-  **Department of Physics Scholarship**, Kent State University, Kent, OH.
- 2022 – 2024  **Graduate Dean's Award**, Kent State University, Kent, OH.
- 2020 – 2021  **Frank Honors Scholarship**, University of Nevada, Las Vegas, NV.
-  **Russell L. Brenda Frank Endowed Scholarship**, University of Nevada, Las Vegas, NV.
- 2020  **NASA Nevada Space Grant**, University of Nevada, Las Vegas, NV.

Certifications/Summer Schools

- 2026  **BNL-CERN School on Physics at Future Colliders - Expanding our Horizons**, European Organization for Nuclear Research, Geneva, CH.
- 2024  **Center for Frontiers in Nuclear Science Summer School - Physics of the Electron-Ion Collider**, Stony Brook University, Stony Brook, NY.
- 2023  **Qiskit Global Summer School 2023 - Quantum Excellence, Theory to Implementation**.
- 2020  **Graduate Research Certification - Grad Rebel Advantage Program**, University of Nevada, Las Vegas, NV.

Collaborations/Working Groups

- 2026 – Present  **H1**, DESY (Deutsches Elektronen-Synchrotron) Lab, Hamburg, Germany.
- 2025 – Present  **ePIC**, Brookhaven National Lab, Upton, NY, USA.
-  **Exclusive/Diffractive/Tagging** Physics Working Group for ePIC, Brookhaven National Lab, Upton, NY, USA.
- 2024 – Present  **STAR**, Brookhaven National Lab, Upton, NY, USA.
-  **EIC Users Group**, Brookhaven National Lab, Upton, NY, USA.

Conference Presentations

- 2026  **Towards Improved Hadron Femtography, Edition V**, "Analysis and Reconstruction Strategies for Exclusive Diffractive eA Physics at ePIC", University of Virginia, Charlottesville, VA.
-  **Nuclear Physics Seminar**, "Imaging Nuclei Through Exclusive Vector Meson Production", Brookhaven National Lab, Upton, NY.
-  **ePIC and EIC Physics Readiness Workshop**, "Incoherent Background Suppression in ϕ Production in $e+A$ Collisions", University of Calabria, Cosenza, IT.
-  **ePIC Collaboration Meeting**, "Diffractive ϕ in eAU Collisions at ePIC", Brookhaven National Lab, Upton, NY.
- 2025  **ePIC and EIC Physics Readiness Workshop**, "Reimagining $|t|$: Advanced Approach to Momentum Transfer Measurement", Institute of Physics, London, UK.
-  **Center for Nuclear Research Seminar**, "From Diffraction to Definition: Imaging Nuclei with Vector Mesons", Kent State University, Kent, OH.
-  **EIC/ePIC Joint Collaboration Meeting**, "Nuclear Imaging Through Coherent Exclusive Vector Meson Production", Jefferson Lab, Newport News, VA.

Conference Presentations (continued)

- **EIC Early Science Workshop**, "Improving $|t|$ Measurement Through Exclusive Coherent Vector Meson Production", Stony Brook University, Stony Brook, NY.
- **Graduate Student Symposium**, "Projective Imaging of Nuclei at the Electron-Ion Collider", Kent State University, Kent, OH.
- **Physics Opportunities at an Electron-Ion Collider**, "Projective Imaging of High-Energy Nuclei Through Exclusive Vector Meson Production in Electron-Ion Collisions", Florida International University, Miami, FL.
- 2024 ■ **Center for Nuclear Research Seminar**, "On Diffractive Physics", Kent State University, Kent, OH.
- **American Physical Society - Eastern Great Lakes Section**, "Investigating Novel Ways of Improving Nuclear Imaging Through Exclusive Vector Meson Production at the Electron-Ion Collider", Marietta University, Marietta, OH.

Publications

- 2026 ■ **The ePIC Exclusive, Diffractive and Tagging Physics Working Group on behalf of the ePIC Collaboration**. Opportunities for Exclusive, Diffractive and Tagging Physics at the Electron-Ion Collider in the Early Science Period. *In preparation*, (2026).
- **ePIC Collaboration**. ePIC Early Science Report. *arXiv:2608.05377*, [nucl-ex], DOI: 10.48550/arXiv.2608.05377.
- **ePIC Collaboration**. The ePIC Detector Preliminary Technical Design Report. Zenodo (2026), DOI: 10.5281/zenodo.18271602.
- **STAR Collaboration**. Energy Correlators within Jets in Transversely Polarized Proton-Proton Collisions at $\sqrt{s_{NN}} = 200$ GeV. *Phys. Rev. Lett.* 137 (2026) 11, 111902, e-Print: 2604.15546 [hep-ex], DOI: 10.1103/fgsf-gx2q.
- **STAR Collaboration**. Inclusive jet cross section in pp collisions at $\sqrt{s_{NN}} = 200$ and 510 GeV. e-Print: 2603.28695 [hep-ex].
- **STAR Collaboration**. Probing Late-Stage Hadronic Interactions at High Baryon Density via K^{*0} Production in the RHIC Beam Energy Scan Program. *Phys. Lett. B* 876 (2026) 140389, e-Print: 2601.14884 [nucl-ex], 10.1016/j.physletb.2026.140389.
- **STAR Collaboration**. Measurement of fifth- and sixth-order fluctuations of (net-)proton number in Au + Au collisions from phase II of the beam energy scan program at RHIC. *Phys. Rev. C* 113 (2026) 5, L051901, e-Print: 2512.19352 [nucl-ex], DOI: 10.1103/lq8c-9fr6.
- **STAR Collaboration**. Identified charged hadron production in Au + Au collisions at $\sqrt{s_{NN}} = 54.4$ GeV with the STAR detector. *Phys. Rev. C* 113 (2026) 5, 054907, e-Print: 2512.06415 [nucl-ex], DOI: 10.1103/4161-dflc.
- **STAR Collaboration**. Evidence of Spin-Interference Effects in Exclusive $J/\psi \rightarrow e^+e^-$ Photoproduction in Ultraperipheral Heavy-Ion Collisions. *Phys. Rev. Lett.* 136 (2026) 24, 242302, e-Print: 2512.02865 [nucl-ex], DOI: 10.1103/tcdb-ldh8.
- **STAR Collaboration**. First Observation of Deuteron- Λ Correlations at RHIC. *Phys. Rev. Lett.* 136 (2026) 24, 242303, e-Print: 2511.15493 [nucl-ex], DOI: 10.1103/3m26-5y83.
- **STAR Collaboration**. Study of Central Exclusive Production of $\pi^+\pi^-$, K^+K^- , and $p\bar{p}$ pairs in proton-proton collisions at $\sqrt{s_{NN}} = 510$ GeV with the STAR detector at RHIC. *JHEP* 07 (2026) 143, e-Print: 2510.27482 [hep-ex], DOI: 10.1007/JHEP07(2026)143.
- **STAR Collaboration**. Measurement of transverse polarization of Λ and $\bar{\Lambda}$ hyperons inside jets in unpolarized proton-proton collisions at $\sqrt{s_{NN}} = 200$ GeV. *JHEP* 06 (2026) 050, e-Print: 2509.17487 [hep-ex], DOI: 10.1007/JHEP06(2026)050.

Publications (continued)

- 2025
- **STAR Collaboration.** Observation of Charmonium Sequential Suppression in Heavy-Ion Collisions at the Relativistic Heavy Ion Collider. *Phys. Rev. Lett.* 136 (2026) 12, 122302, e-Print: 2509.12842 [nucl-ex], DOI: 10.1103/ghzh-kv8z.
 - **STAR Collaboration.** Measurement of inclusive J/ψ production in Au+Au collisions at $\sqrt{s_{NN}} = 54.4$ GeV at STAR. *Phys. Lett. B* 877 (2026) 140405, e-Print: 2506.20962 [nucl-ex], DOI: 10.1016/j.physletb.2026.140405.
 - **STAR Collaboration.** Measuring spin correlation between quarks during QCD confinement. *Nature* 650 (2026) 8100, 66-71, e-Print: 2506.05499 [hep-ex], DOI: 10.1038/s41586-025-09920-0.
 - **M. Kesler, A. I. Sheikh, R. Ma, Z. Tu, T. Ullrich, and Z. Xu.** Projective imaging of high-energy nuclei via coherent exclusive vector meson production in electron-nucleus collisions. *Phys. Lett. B* 875 (2026) 140585, e-Print: 2502.15596 [nucl-th], DOI: 10.1016/j.physletb.2026.140585.
 - **STAR Collaboration.** Measurement of the longitudinal flow-plane decorrelation using multi-plane cumulants in $\sqrt{s_{NN}} = 200$ GeV Au + Au, Ru + Ru, and Zr + Zr collisions. *Phys. Rev. C* 113 (2026) 2, 024908, e-Print: 2506.02330 [nucl-ex], DOI: 10.1103/75tf-ljtk.
 - **STAR Collaboration.** Search for the chiral magnetic effect through beam energy dependence of charge separation using event shape selection. *Phys. Rev. C* 113 (2026) 1, 014912, e-Print: 2506.00278 [nucl-ex], DOI: 10.1103/pj7s-9ym7.
 - **STAR Collaboration.** Measurement of medium-induced acoplanarity in central Au-Au and pp collisions at $\sqrt{s_{NN}} = 200$ GeV using direct-photon+jet and π^0 + jet correlations. *Phys. Rev. C* 113 (2026) 1, 014902, e-Print: 2505.05789 [nucl-ex], DOI: 10.1103/k29c-d5ry.
 - **M. Kesler, A. I. Sheikh, R. Ma, Z. Tu, T. Ullrich, and Z. Xu.** ePIC Analysis Note: Precision Extraction of Momentum Transfer in Diffractive Coherent Exclusive Vector Meson Production. Zenodo (2025), DOI: 10.5281/zenodo.17944278.
 - **STAR Collaboration.** Energy independence of the Collins asymmetry in p↑p collisions. *Phys. Rev. Lett.* 135 (2025) 26, 261902, e-Print: 2507.16355 [hep-ex], DOI: 10.1103/jgsw-zhny.
 - **STAR Collaboration.** Hyperon global polarization in isobar Ru+Ru and Zr+Zr collisions at $\sqrt{s_{NN}}=200$ GeV. *Phys. Lett. B* 870 (2025), 139891, e-Print: 2505.05046 [nucl-ex], DOI: 10.1016/j.physletb.2025.13989.
 - **STAR Collaboration.** Onset of constituent quark number scaling in heavy-ion collisions at RHIC. *Phys. Rev. Lett.* 135 (2025) 7, 072301, e-Print: 2504.02531 [nucl-ex], DOI: 10.1103/2qhx-cp79.
 - **STAR Collaboration.** Precision measurement of net-proton-number fluctuations in Au+Au collisions at RHIC. *Phys. Rev. Lett.* 135 (2025) 14, 142301, e-Print: 2504.00817 [nucl-ex], DOI: 10.1103/9l69-2d7p.
 - **STAR Collaboration.** Measurement of two-point energy correlators within jets in p+p Collisions at $\sqrt{s}=200$ GeV. *Phys. Rev. Lett.* 135 (2025) 11, 111901, e-Print: 2502.15925 [hep-ex], DOI: 10.1103/wv2t-dkgn.
 - **STAR Collaboration.** Measurements of ψ states production in p+p collisions at $\sqrt{s}=500$ GeV with STAR: Cross sections, ratios, and multiplicity dependence. *Phys. Rev. D* 112 (2025) 3, 032004, e-Print: 2502.03769 [hep-ex], DOI: 10.1103/bsyx-qtjp.
 - **STAR Collaboration.** Temperature measurement of Quark-Gluon plasma at different stages. *Nature Commun.* 16 (2025) 1, 9098, e-Print: 2402.01998 [nucl-ex], DOI: 10.1038/s41467-025-63216-5.